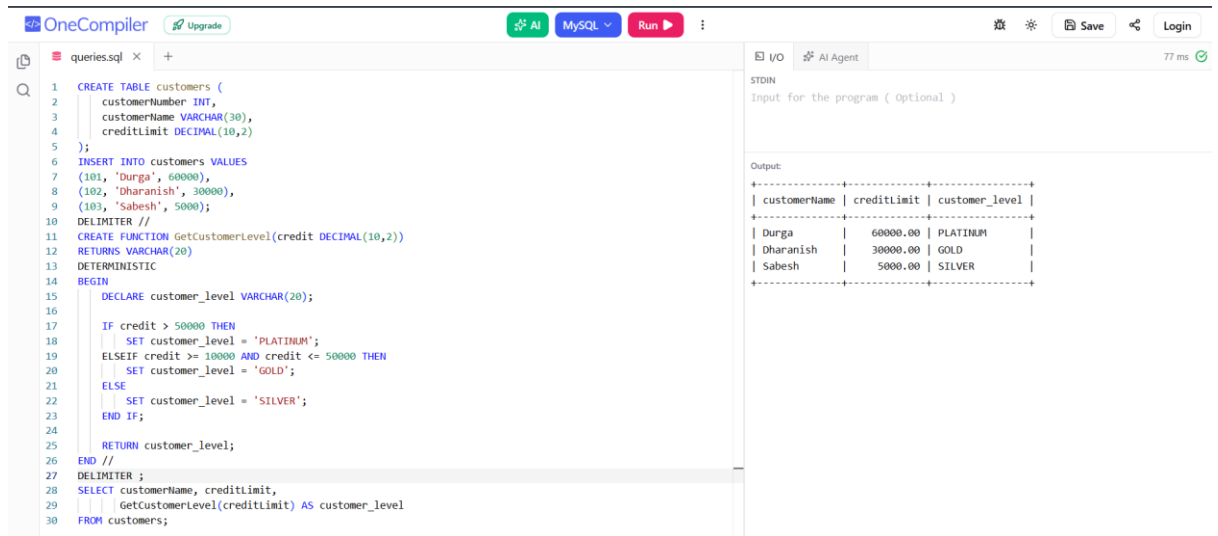


Functions

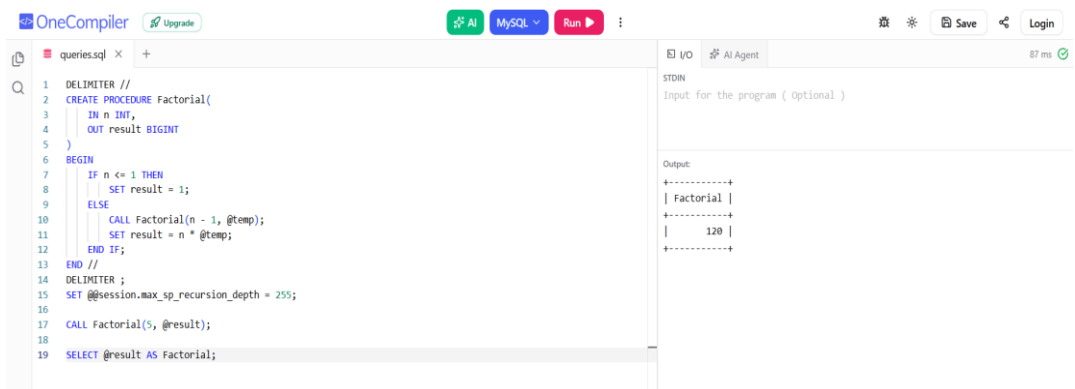


The screenshot shows the OneCompiler MySQL IDE. The left pane contains a SQL script named 'queries.sql'. The script creates a 'customers' table with columns 'customerNumber' (INT), 'customerName' (VARCHAR(30)), and 'creditLimit' (DECIMAL(10,2)). It inserts three records: (101, 'Durga', 60000), (102, 'Dharanish', 30000), and (103, 'Sabesh', 5000). Then, it creates a function 'GetCustomerLevel' that takes a 'credit' parameter and returns a 'customer_level' (VARCHAR(20)). The function uses a series of IF-ELSE statements to assign levels: 'PLATINUM' for credit > 50000, 'GOLD' for credit >= 10000 and <= 50000, and 'SILVER' for credit <= 5000. Finally, it selects all customer data and applies the 'GetCustomerLevel' function to the 'creditLimit' column, aliasing the result as 'customer_level'.

```
1 CREATE TABLE customers (  
2     customerNumber INT,  
3     customerName VARCHAR(30),  
4     creditLimit DECIMAL(10,2)  
5 );  
6 INSERT INTO customers VALUES  
7 (101, 'Durga', 60000),  
8 (102, 'Dharanish', 30000),  
9 (103, 'Sabesh', 5000);  
10 DELIMITER //  
11 CREATE FUNCTION GetCustomerLevel(credit DECIMAL(10,2))  
12 RETURNS VARCHAR(20)  
13 DETERMINISTIC  
14 BEGIN  
15     DECLARE customer_level VARCHAR(20);  
16  
17     IF credit > 50000 THEN  
18         SET customer_level = 'PLATINUM';  
19     ELSEIF credit >= 10000 AND credit <= 50000 THEN  
20         SET customer_level = 'GOLD';  
21     ELSE  
22         SET customer_level = 'SILVER';  
23     END IF;  
24  
25     RETURN customer_level;  
26 END //  
27 DELIMITER ;  
28 SELECT customerName, creditLimit,  
29        GetCustomerLevel(creditLimit) AS customer_level  
30 FROM customers;
```

The right pane shows the output of the query. It displays a table with three columns: 'customerName', 'creditLimit', and 'customer_level'. The data rows are: Durga (60000.00, PLATINUM), Dharanish (30000.00, GOLD), and Sabesh (5000.00, SILVER).

customerName	creditLimit	customer_level
Durga	60000.00	PLATINUM
Dharanish	30000.00	GOLD
Sabesh	5000.00	SILVER



The screenshot shows the OneCompiler MySQL IDE. The left pane contains a SQL script named 'queries.sql'. The script creates a recursive procedure 'Factorial' that takes an integer 'n' and returns a result. The procedure uses a recursive call to calculate the factorial of 'n-1' and then multiplies it by 'n'. It also sets a session variable 'max_sp_recursion_depth' to 255. Finally, it calls the 'Factorial' procedure with the value 5 and selects the result.

```
1 DELIMITER //  
2 CREATE PROCEDURE Factorial(  
3     IN n INT,  
4     OUT result BIGINT  
5 )  
6 BEGIN  
7     IF n <= 1 THEN  
8         SET result = 1;  
9     ELSE  
10        CALL Factorial(n - 1, @temp);  
11        SET result = n * @temp;  
12    END IF;  
13 END //  
14 DELIMITER ;  
15 SET @@session.max_sp_recursion_depth = 255;  
16  
17 CALL Factorial(5, @result);  
18  
19 SELECT @result AS Factorial;
```

The right pane shows the output of the query. It displays a single row with the value 120, which is the factorial of 5.

Factorial
120